Output Ontology (OO)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *MILU* | ***Context:*** *North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output ontology is well-aligned with the deployment. Ground-truth labels are single correct answers per MCQ drawn from exam answer keys [Q12], which exactly matches the deployment's binary correct/incorrect requirement with positive/negative marking. The label schema is binary at inference time, and the elicitation explicitly confirmed that 'evaluation schemas across boards and states are described as broadly similar' [A4], so competitive-exam keyed answers are considered compatible. The 41-subject 8-domain organisation [Q42] supports per-subject reporting useful for diagnostic analysis. Domain-level analysis surfaces a known weakness of LLMs in culturally specific areas [Q20, Q71, Q77], which the deployment user should be aware of. The taxonomy was derived bottom-up from ~20K fine-grained tags clustered and manually merged [Q39, Q40, Q41], which is sound methodology, although whether the resulting categories carry any culturally variable correct-answer norms is not analysed.

ID	Evidence
Q12	“These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science.” (p.2)
Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
Q20	“Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM.” (p.2)
Q71	“We analyze the performance of various base and instruct models across multiple domains and languages. Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields.” (p.7)
Q77	“Additionally, the domain-specific analysis indicates that models perform better in general fields such as STEM while facing challenges in culturally relevant subjects like Arts, Humanities, and Law, highlighting the lack of this knowledge in the current models and datasets.” (p.8)
Q39	“Finally, in total, we get around 20K tags. However, these tags are highly fine-grained, often having a heavy overlap.” (p.4)
Q40	“To organize them, we embed the tags using the NV-EMBED-V2 (Lee et al., 2024) model and apply K-means clustering to group tags into 50 clusters.” (p.4)
Q41	“We manually review these clusters and assign appropriate subject labels.” (p.4)

Strengths

- Single-correct-answer ground truth from exam answer keys directly matches binary scoring requirement [Q12]
- 41-subject 8-domain organisation supports diagnostic per-subject reporting [Q42]
- Cross-language answer-key consistency for parallel items [DATASET-D18, D19] confirms reliable label application
- Elicitation confirms competitive-exam keys are broadly compatible with North Indian board norms [A4]

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Q12	“These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science.” (p.2)

Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D19	1991 में वित्तीय संकट को ट्रगिर करने वाले वदिेशी मुद्रा भंडार की हानि का तात्कालकि कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are MCQ option indices (option1–option4) with a single target [Q12, DATASET-D1]. This is regionally relevant — Indian competitive and board MCQs use the same single-correct format with positive/negative marking [WEB-1, WEB-2].

OO-2: No regionally specific output categories appear missing for a binary MCQ deployment. The deployment does not require partial-credit, rubric, or multi-label output [A1, A2].

OO-3: No categories visibly encode non-regional values. The user confirmed evaluation schemas are broadly consistent across North Indian boards and competitive exams [A4], reducing the risk that competitive-exam answer keys diverge from board norms.

OO-4: Stakeholder-driven taxonomy redesign is not required — the deployment uses the same single-answer MCQ taxonomy as the benchmark.

OO-5: No major taxonomy issues identified. Minor caveat: domain-level categorisation may aggregate over heterogeneous content (e.g., Arts & Humanities mixing competitive-GK and literature) [Q20, Q77], which could affect interpretation of domain-level scores.

ID	Evidence
Q12	“These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science.” (p.2)
WEB-1	Indian competitive exams use single-correct MCQ with positive/negative marking — taxonomically aligned with MILU output schema (https://negativemarkingcalculator.in/)
WEB-2	UPSC and similar exams confirm single-answer MCQ output convention (https://testbook.com/upsc-civil-services/negative-marking-in-upsc)
D1	जब एक डीसी सीरीज मोटर बिना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering question about DC series motor — translated from English
Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
Q20	“Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM.” (p.2)
Q77	“Additionally, the domain-specific analysis indicates that models perform better in general fields such as STEM while facing challenges in culturally relevant subjects like Arts, Humanities, and Law, highlighting the lack of this knowledge in the current models and datasets.” (p.8)

Information Gaps

- Whether domain-level aggregation hides subject-level mismatches relevant to the teacher deployment

Requires Expert Verification

- Whether competitive-exam answer keys for items in Hindi Literature and Civics agree with state-board mark-scheme conventions in detail